

A1 和差化積

令 $\theta = \omega t$: 把 X, Y 都改寫成餘弦

$$a = \theta - \frac{\pi}{3}, b = \theta + \frac{\pi}{6}$$

$$\sin(\theta + \frac{2}{3}\pi) = \cos(\theta + \frac{\pi}{6}) = \cos b$$

$$X = \sqrt{3} \cos a, Y = 3 \cos b$$

$$\cos a + \cos b = 2 \cos(\frac{a+b}{2}) \cdot \cos(\frac{a-b}{2})$$

$$= \sqrt{2} \cos(\frac{\theta - \pi}{12})$$

$$\cos a - \cos b = -2 \sin(\frac{a+b}{2}) \cdot \sin(\frac{a-b}{2})$$

$$= \sqrt{2} \sin(\frac{\theta - \pi}{12})$$

$$Z = \frac{\sqrt{3}+3}{2} (\cos a + \cos b) + \frac{\sqrt{3}-3}{2} (\cos a - \cos b)$$

$$= \frac{\sqrt{3}+3}{\sqrt{2}} \cos(\frac{\theta - \pi}{12}) + \frac{\sqrt{3}-3}{\sqrt{2}} \sin(\frac{\theta - \pi}{12})$$

$$\frac{\sqrt{3}+3}{\sqrt{2}} = 2\sqrt{3} \cos(\frac{\pi}{12})$$

$$\frac{\sqrt{3}-3}{\sqrt{2}} = -2\sqrt{3} \sin(\frac{\pi}{12})$$

$$Z = 2\sqrt{3} [\cos(\frac{\pi}{12}) \cos(\frac{\theta - \pi}{12}) - \sin(\frac{\pi}{12}) \sin(\frac{\theta - \pi}{12})]$$

$$= 2\sqrt{3} \cos \theta = 2\sqrt{3} \cos(\omega t) \quad \#$$

A₂ 相子

$$\theta = \omega t$$

$$Y(t) = 3 \sin(\theta + \frac{2\pi}{3}) = 3 \cos(\theta + \frac{\pi}{6})$$

$$X = \sqrt{3} e^{-\frac{j\pi}{3}} = \frac{\sqrt{3}}{2} - \frac{3j}{2}$$

$$Y = 3 e^{\frac{j\pi}{6}} = \frac{3\sqrt{3}}{2} + \frac{3}{2}j$$

$$Z = X + Y = 2\sqrt{3} + j0$$

$$Z(t) = \operatorname{Re}[Z e^{j\omega t}]$$

$$= 2\sqrt{3} \cos(\omega t)$$

A3 Geo Gebra 圖形板對

令 $x = \omega t$, 下列 f, g, h 分別對應 X, Y, Z

$$f(x) = \sqrt{3} \cos(x - \frac{\pi}{3})$$

$$g(x) = 3 \sin(x + \frac{2\pi}{3})$$

$$h(x) = f(x) + g(x) = 2\sqrt{3} \cos x$$

$$x=0 : f(0) = \frac{\sqrt{3}}{2}, g(0) = \frac{3\sqrt{3}}{2}$$

$$h(0) = 2\sqrt{3} \approx 3.464 : h \text{ 相位為 } 0$$

B1 連續時間穩態

令 $T = RC$: 由 KVL 得電路微分方程

$$x(t) = T \frac{dy(t)}{dt} + y(t)$$

$$x(t) = e^{j\omega t} \quad \text{設 } y_s(t) = H(\omega) e^{j\omega t}$$

$$1 = (1 + j\omega T) H(\omega)$$

$$H(\omega) = \frac{1}{1 + j\omega T}$$

$$H(\omega) = \frac{1 + j\omega T}{(1 + j\omega T)^2}$$

$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega T)^2}}$$

$$\phi(\omega) = -\tan^{-1}(\omega T)$$

$$\text{實數餘弦輸入} = y_s(t) = |H| \cos(\omega t + \phi)$$

B₂ 暫態與穩態

$$X(t) = e^{j\omega t} u(t), \text{ 假設 } y(0^-) = 0$$

$$t > 0: T - \frac{dy}{dt} + y = e^{j\omega t}$$

$$\text{特解: } y_p(t) = H(\omega) e^{j\omega t}$$

$$\text{齊次解: } y_h(t) = K e^{\frac{-t}{T}}$$

$$\text{總解: } y(t) = H(\omega) e^{j\omega t} + K e^{\frac{-t}{T}}$$

$$y(0^+) = 0 \Rightarrow H(\omega) + K = 0 \Rightarrow K = -H(\omega)$$

$$y(t) = H(\omega) \cdot (e^{j\omega t} - e^{\frac{-t}{T}}) \cdot u(t)$$

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B3 RC 參數與連續響應

$$R = 1000 \Omega, C = \frac{1}{2\pi \cdot 400 \cdot 1000} \text{ F}$$

$$C = 3.97887358 \times 10^{-7} \text{ F}$$

$$T = RC = 3.97887358 \times 10^{-4} \text{ s} \approx 0.398 \text{ ms}$$

$$f_c = \frac{1}{2\pi RC} = 400 \text{ Hz}$$

$$\omega = 2\pi f \Rightarrow \omega T = \frac{f}{400}$$

$$H(f) = \frac{400}{400 + jf}$$

B4 三種頻率的輸出

$$y_{f_{\text{ext}}} = |H(f)| \cos(2\pi f t + \phi_f) - \operatorname{Re} \cdot H(f) \cdot e^{\frac{-t}{T}}$$

$$T = 3.97887358 \times 10^{-4} \text{ s}, t \geq 0$$

$$f = 100 \text{ Hz}$$

$$y_{100}(t) = 0.970143 \cos(200\pi t - 14.036^\circ) - 0.941176 e^{\frac{-t}{T}}$$

$$f = 400 \text{ Hz}$$

$$y_{400}(t) = 0.107107 \cos(800\pi t - 45^\circ) - 0.5 e^{\frac{-t}{T}}$$

$$f = 3000 \text{ Hz}$$

$$y_{3000}(t) = 0.132164 \cos(6000\pi t - 82.405^\circ) - 0.017467 e^{\frac{-t}{T}}$$

B5 離散式與頻率響應

$$T = \frac{1}{f_s}, \quad \frac{dy}{dt} = \frac{y(n) - y(n-1)}{T}$$

$$x(n) = \frac{T}{\tau} \cdot [y(n) - y(n-1)] + y(n)$$

$$(T + \tau)y(n) = T \cdot y(n-1) + \tau x(n)$$

$$y(n) = a y(n-1) + B x(n)$$

$$a = \frac{\tau}{T + \tau}, \quad B = \frac{T}{T + \tau} = 1 - a$$

$$x(n) = e^{j\omega n}, \quad \omega = \frac{2\pi f}{f_s}$$

$$\text{設 } y(n) = |H_d(e^{j\omega})| e^{j\omega n}$$

$$H_d(e^{j\omega}) = \frac{B}{1 - a e^{-j\omega}}$$

$$|H_d| = \frac{B}{\sqrt{(1 - a \cos \omega)^2 + (a \sin \omega)^2}}$$

$$\angle_d = -a \tan 2 \angle(a \sin \omega, 1 - a \cos \omega)$$

B6 不同取樣率比較

$$T = \frac{1}{8000}, \alpha = \frac{T \cdot f_s}{T+1}$$

連續RL(B3-B4):

$$100 \text{ Hz} = \frac{0.970143}{-14.036^\circ}$$

$$400 \text{ Hz} = \frac{0.967107}{-45^\circ}$$

$$3000 \text{ Hz} = \frac{0.132164}{-82.405^\circ}$$

$$f_s = 4000 \text{ Hz}, \alpha = 0.614130$$

$$100 \text{ Hz} = \frac{0.952787}{-13.1722^\circ}$$

$$400 \text{ Hz} = \frac{0.623123}{-35.657^\circ}$$

$$3000 \text{ Hz} = \frac{0.328813}{31.555^\circ}$$

$$f_s = 8000 \text{ Hz}, \alpha = 0.760943$$

$$100 \text{ Hz} = \frac{0.96132}{-13.891^\circ}$$

$$400 \text{ Hz} = \frac{0.658896}{-40.399^\circ}$$

$$3000 \text{ Hz} = \frac{0.146709}{-19.282^\circ}$$

$$f_s = 16000 \text{ Hz}, \alpha = 0.864245$$

$$100 \text{ Hz} = \frac{0.965695}{-13.967^\circ}$$

$$400 \text{ Hz} = \frac{0.68125}{-42.423^\circ}$$

$$3000 \text{ Hz} = \frac{0.13032}{-50.030^\circ}$$

比較: $f_s = 4000 \text{ Hz}$ 時, $f_N = 2000 \text{ Hz}$; 3000 Hz 取樣後產生混疊成 -1000 Hz
 因此這筆結果不能直接與連續時間 3000 Hz 結果比較